

Humza Inam . Jarvis Consulting

Computer Engineering graduate with hands-on experience in data engineering, analytics, and cloud infrastructure. Completed Python analytics, Databricks ETL/DLT pipelines, Linux monitoring, SQL analysis, and Java development projects at Jarvis. Proficient in Java, Python, SQL (PostgreSQL), PySpark, and cloud platforms (Azure, AWS, Databricks). Experienced with ETL pipelines, Delta Lake, real-time streaming, and serverless architectures. Strong foundation in object-oriented programming, data analysis (Pandas/NumPy), and agile methodologies. Seeking a Data Engineering or Analytics role to leverage technical expertise in building scalable data solutions and delivering actionable business insights.

Skills

Proficient: Java, Python, SQL (PostgreSQL/MySQL), Linux/Bash, Git, Docker

Competent: Pandas/NumPy/Jupyter Notebook, Databricks, PySpark, AWS (Lambda, Bedrock, API Gateway, S3, DynamoDB), Azure (Event Hub, Stream Analytics, Functions, SQL, ADLS, Blob Storage), ETL/Data Pipelines, Delta Lake, REST APIs, JavaScript/TypeScript, React.js/Next.js/Angular, Express.js/.NET, MongoDB/Supabase/Firebase

Familiar: Terraform/IaC, C/C#, Power BI, Jest/Jasmine/Unit Testing, CI/CD (Vercel), Microservices Architecture, HTML/CSS/TailwindCSS, Scikit-learn

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_HumzaInam

Home Credit Default Scoring Risk ML [GitHub]: Built an end-to-end machine learning credit risk scoring system using Python, Pandas, NumPy, and Scikit-learn. Performed extensive exploratory data analysis (EDA), missing data analysis, and feature engineering on the Home Credit dataset. Engineered predictive features including aggregated bureau-level credit history signals. Trained and evaluated multiple models including Logistic Regression, Random Forest, XGBoost, and LightGBM using AUC, Gini, KS, and AUPRC metrics. Applied class imbalance handling using `scale_pos_weight` and `class_weight` techniques. Identified XGBoost as the best-performing model and optimized decision thresholds for credit default prediction.

London Gift Shop Customer Analytics [GitHub]: Developed a customer analytics proof-of-concept for a UK-based e-commerce retailer using Python, Jupyter Notebook, Pandas, and NumPy to analyze transactional retail data and customer purchasing behavior. Performed data cleaning, validation, and feature engineering to transform raw invoice-level data into customer-level insights. Implemented RFM (Recency, Frequency, Monetary) analysis and customer segmentation to identify high-value, loyal, and at-risk customers. Generated monthly user activity metrics and business-ready summaries to support targeted marketing campaigns and customer retention strategies for the LGS marketing team.

Databricks ETL & DLT Pipelines [GitHub]: Built two data engineering pipelines on Databricks using medallion architecture (Bronze, Silver, Gold). Developed fraud detection ETL pipeline processing credit card transaction data from Azure SQL Database, Blob Storage, and ADLS Gen2 using JDBC, Azure Data Factory, and Lakeflow Connect. Implemented data transformation with PySpark, creating 15 analytics tables to identify fraud patterns by user behavior, merchant categories, transaction timing, and spending anomalies. Designed automated Databricks Workflows with scheduled job orchestration and published SQL dashboards with interactive visualizations. Built Delta Live Tables pipeline for stock market analytics, ingesting daily data from Alpha Vantage API, applying data quality checks, and generating aggregated metrics. Applied Delta Lake for ACID transactions, designed complex joins and aggregations, and implemented end-to-end data pipelines from ingestion to business intelligence reporting.

Linux Cluster Resource Monitoring App [GitHub]: Developed a Linux cluster monitoring solution using Bash scripts and PostgreSQL to track hardware specifications and real-time resource usage across Rocky Linux servers. Implemented Docker containerization for PostgreSQL database, automated data collection via crontab, and designed SQL queries for resource analytics. Collected CPU, memory, and disk metrics every minute to identify low-memory servers and optimize capacity planning. Applied GitFlow version control and deployed on Google Cloud Platform for the Jarvis LCA team.

Java Grep Application [GitHub]: Built a Java-based command-line grep application to recursively search directory trees for regex pattern matches and output results to a file. Implemented file system traversal, buffered file I/O, and regex matching using core Java APIs. Applied object-oriented design principles with interface-based abstraction, exception handling, and logging via SLF4J and Log4j. Designed the application for scalability and maintainability, supporting large directory structures and automated execution in Linux environments.

Highlighted Projects

AML RAG Agent (AWS Bedrock + Lambda + Knowledge Base) [GitHub]: Built a serverless Anti-Money Laundering (AML) assistant using AWS Bedrock foundation models, AWS Knowledge Bases for Retrieval-Augmented Generation (RAG), LangChain/LangGraph agents, and AWS Lambda (containerized via SAM). Implemented tool-based workflows for transaction risk scoring, simulated sanctions checks, document retrieval from S3, and structured compliance report generation, exposed through API Gateway with an optional Streamlit UI. Designed the system to ground model responses in indexed compliance documents. Developed as a hands-on learning project to explore LLM agents, RAG architecture.

IoT Real-time Azure Streaming Pipeline [GitHub]: Engineered a real-time IoT data pipeline with machine learning anomaly detection, streaming telemetry from Azure IoT Central through Event Hub and Stream Analytics into Azure SQL Database and Azure Data Lake Storage. Processed 8000+ device telemetry points with sub-second latency using Event Hub ingestion and real-time data enrichment. Developed Azure Functions with timer triggers to incrementally fetch SQL telemetry data and push to Power BI API for visualization dashboards. Integrated .NET 8.0 Azure Functions for serverless compute and automated data transformation workflows. Demonstrated end-to-end ETL pipeline architecture with cloud-native services for IoT data processing, storage, and business intelligence reporting.

Azure Networking Environment [GitHub]: Engineered a secure Azure cloud networking environment using Terraform Infrastructure as Code, provisioning VNets, subnets, NAT Gateway, Private Endpoints, VNet peering, and Application Gateway for load-balanced multi-tier connectivity. Automated deployment of Spring Boot applications on private VMs using IaC best practices, eliminating manual configuration errors. Validated DNS resolution, private storage access, and cross-VNet communication via REST APIs to ensure secure data flow. Demonstrated enterprise-grade cloud architecture patterns including network isolation, traffic routing, and security boundaries. Implemented declarative infrastructure management enabling version-controlled, reproducible deployments across multiple environments.

Mindryx - AI Quiz Generation Study Platform [GitHub]: Available at <https://mindryxify.vercel.app/> - Built a full-stack serverless quiz platform with AI-generated quizzes from topics or PDFs using Gemini API. Developed with Next.js frontend and Supabase backend featuring PostgreSQL database, Edge Functions for serverless compute, and Clerk authentication. Implemented LocalStack for local development to emulate AWS Lambda, DynamoDB, and SQS, then migrated to production using Supabase Edge Functions for data processing workflows. Integrated Stripe payment gateway for subscription management and implemented PDF OCR using Tesseract.js for document processing. Deployed CI/CD pipeline via Vercel with automated testing and privacy-focused in-browser WebLLM chat functionality. Designed scalable database schema and RESTful APIs to handle user data, quiz generation, and analytics tracking.

SafeScript - VS Code Extension Capstone Project [GitHub]: Led UI development for an award-winning VS Code extension using Abstract Syntax Tree (AST) static analysis to detect buffer overflows and critical C vulnerabilities with optimized real-time analysis under 3 seconds. Designed Security Analysis panels, AI Suggestion History interface, and integrated Code Llama for automated code generation and vulnerability remediation. Developed static analysis rules with CWE/CVE mapping for industry-standard security classifications and comprehensive Jest unit tests for quality assurance. Managed Agile development sprints via Jira with iterative feature delivery and stakeholder feedback integration.

Professional Experiences

Data Engineer, Jarvis Consulting Group (Nov 2025 - Present): Developing data engineering and analytics solutions using Java, Python, SQL, and cloud technologies in an Agile environment. Completed Linux cluster monitoring system with Bash scripting, PostgreSQL database design, and automated data collection via cron jobs for infrastructure resource tracking. Built comprehensive SQL database management system demonstrating complex queries, data modeling, and business intelligence analytics. Developed Core Java grep application with object-oriented design, file I/O operations, and regex pattern matching for automated text processing. Created Python-based customer analytics proof-of-concept using Pandas and NumPy for retail data analysis, implementing RFM segmentation and generating actionable business insights for marketing teams. Built end-to-end Databricks pipelines including fraud detection ETL with Azure integration (SQL Database, Blob Storage, ADLS Gen2) and Delta Live Tables for stock market analytics. Implemented medallion architecture (Bronze, Silver, Gold), PySpark transformations, Delta Lake storage, automated workflows, and SQL dashboards for business intelligence reporting. Collaborating in Agile teams using Git version control, Scrum methodologies, and Docker containerization for reproducible deployments.

Software Developer Intern, SOTI Inc. (Sep 2023 - Aug 2024): Resolved 130+ issues, including bugs, feature enhancements, and unit tests, over a 10-month period, significantly improving the frontend user experience in localization, aesthetics, and design, while collaborating closely with QA engineers and Product Owners to ensure seamless delivery. Contributed to upgrading the backend to .NET 8.0, resolving critical security vulnerabilities, and increasing coverage of a

large software module (15,000 lines) from 5% to 80% by writing comprehensive unit tests with Karma and Jasmine, with a small team of developers, enhancing reliability and maintainability across the system.

Digital Transformation Project Lead, Airbond Travel (Jun 2025 - Nov 2025): Integrated a B2C AI trip-planning and booking engine, researched and implemented cost-effective payment gateways, improved accountant payment tracking efficiency by 50%, and developed IT training materials for digital literacy. Advised the President on business technology initiatives, helping shape the agency's digital transformation strategy.

Education

York University (Sep 2020 - Oct 2025), Bachelor of Engineering, Computer Engineering - GPA: 7.38/9.0 - Relevant Coursework: Database Systems, Data Structures & Algorithms, Operating Systems, Software Design, Advanced OOP, Embedded Systems, Robotics, Engineering Capstone

Miscellaneous

- P.E.O. York Chapter Award for Greatest Commercial Potential (Apr 2025): Selected from 50+ capstone teams for SafeScript VS Code extension